Multiplying Fractions and Mixed Numbers
Answer Key

Grade 4–5 Fractions

Quick Answers

1. $\frac{2}{5}$
2. $\frac{3}{8}$
3. $\frac{10}{3}$
4. $\frac{1}{12}$
5. $\frac{3}{2}$
6. $\frac{3}{5}$
7. 3
8. $\frac{9}{2}$
9. $\frac{1}{4}$
10. 4.125
11. $\frac{8}{5}$
12. $\frac{3}{4}$

Curriculum

Level: Grade 4–5 Fractions
5.NF.A.1 / 6.NS.A.1 / 5.NF.B – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

10. *If they got 2.375: multiplied whole parts and fraction parts separately instead of converting to improper fractions*

1. $\frac{2}{3} \times \frac{3}{5}$. Multiply straight across: numerator $2 \times 3 = 6$, denominator $3 \times 5 = 15$, giving $\frac{6}{15}$. Both parts share the factor 3, so divide top and bottom by 3. $\boxed{\frac{2}{5}}$

2. $\frac{3}{4} \times \frac{1}{2}$. Across the top: $3 \times 1 = 3$. Across the bottom: $4 \times 2 = 8$. The numerator 3 and the denominator 8 share no factor, so it is already simplest. $\boxed{\frac{3}{8}}$

3. $\frac{5}{6} \times 4$. Write the whole number as $\frac{4}{1}$ so both factors are fractions: $\frac{5}{6} \times \frac{4}{1} = \frac{20}{6}$. Divide top and bottom by 2. $\boxed{\frac{10}{3}}$ (the same as $3\frac{1}{3}$).

4. $\frac{3}{8} \times \frac{2}{9}$. Cancel first: the 3 on top and the 9 below share 3, leaving $\frac{1}{8} \times \frac{2}{3}$; the 2 on top and the 8 below share 2, leaving $\frac{1}{4} \times \frac{1}{3}$. Now multiply across. $\boxed{\frac{1}{12}}$

5. $2\frac{1}{2} \times \dfrac{3}{5}$. A mixed number must become an improper fraction first: $2\frac{1}{2} = \dfrac{2 \times 2 + 1}{2} = \dfrac{5}{2}$. Then $\dfrac{5}{2} \times \dfrac{3}{5}$ — cancel the 5s — $= \dfrac{3}{2}$. $\boxed{\dfrac{3}{2}}$

6. $\dfrac{3}{4} \times x = \dfrac{9}{20}$. The missing factor is found by dividing: $x = \dfrac{9}{20} \div \dfrac{3}{4} = \dfrac{9}{20} \times \dfrac{4}{3}$. Cancel 3 into 9 and 4 into 20: $\dfrac{3}{5} \times \dfrac{1}{1}$. Check: $\dfrac{3}{4} \times \dfrac{3}{5} = \dfrac{9}{20}$ ✓ $\boxed{x = \dfrac{3}{5}}$

7. $1\frac{1}{3} \times 2\frac{1}{4}$. Convert both: $1\frac{1}{3} = \dfrac{4}{3}$ and $2\frac{1}{4} = \dfrac{9}{4}$. Multiply: $\dfrac{4}{3} \times \dfrac{9}{4}$. The 4s cancel and 3 divides into 9, leaving $\dfrac{1}{1} \times \dfrac{3}{1}$. $\boxed{3}$

8. 12 batches, each using $\dfrac{3}{8}$ of a cup. "Each" with a fraction means multiply: $12 \times \dfrac{3}{8} = \dfrac{12}{1} \times \dfrac{3}{8} = \dfrac{36}{8}$. Divide top and bottom by 4. $\boxed{\dfrac{9}{2}}$ cups (that is $4\frac{1}{2}$ cups).

9. $\dfrac{7}{10} \times \dfrac{5}{14}$. Cancel before multiplying: 7 divides into 14, and 5 divides into 10, leaving $\dfrac{1}{2} \times \dfrac{1}{2}$. Multiplying across gives the answer. $\boxed{\dfrac{1}{4}}$

10. Area of a rectangle is length times width, so the area is $2\frac{3}{4} \times 1\frac{1}{2}$. Convert both to improper fractions: $\dfrac{11}{4} \times \dfrac{3}{2} = \dfrac{33}{8}$. As a decimal, $33 \div 8 = 4.125$. $\boxed{4.125 \text{ ft}^2}$
Watch for the common error: multiplying the whole parts and the fraction parts separately gives 2.375, which is far too small — the rectangle is wider than 2 ft and taller than 1 ft, so its area must exceed 2 square feet by more than that.

11. $x \times 2\frac{1}{2} = 4$. Rewrite the mixed number: $2\frac{1}{2} = \dfrac{5}{2}$, so $x \times \dfrac{5}{2} = 4$. Divide by $\dfrac{5}{2}$, which means multiplying by its reciprocal: $x = 4 \times \dfrac{2}{5} = \dfrac{8}{5}$. Check: $\dfrac{8}{5} \times \dfrac{5}{2} = \dfrac{40}{10} = 4$ ✓ $\boxed{x = \dfrac{8}{5}}$

12. Saturday: Maya walks $\dfrac{2}{3}$ of the trail, so the part left over is $1 - \dfrac{2}{3} = \dfrac{1}{3}$ of the trail. The trail is $4\frac{1}{2} = \dfrac{9}{2}$ kilometres, so the leftover part is $\dfrac{9}{2} \times \dfrac{1}{3} = \dfrac{3}{2}$ km. Sunday she walks half of that leftover piece: $\dfrac{3}{2} \times \dfrac{1}{2} = \dfrac{3}{4}$. $\boxed{\dfrac{3}{4} \text{ km}}$